Titanic — Exploratory Data Analysis

Project 2 Report

Part 1: Data Understanding

This dataset covers 891 Titanic passengers across 12 original columns, including who they were (Name, Sex, Age), how they travelled (Passenger Class, Fare, Cabin, Embarkation), and whether they survived. There are no missing values and no duplicate rows in the cleaned data.

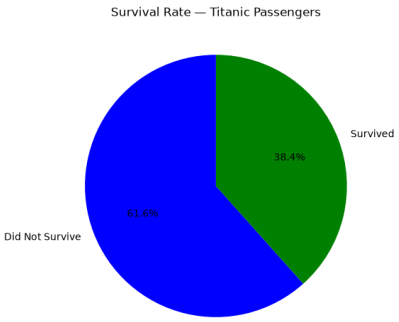
Part 2: Feature Engineering

Feature	What it does
Family Size	Sibling/Spouses Aboard + Parents/Children Aboard + 1 (the passenger). Gives the total size of the travel group.
Is Alone	'No' if Family Size is 1, otherwise 'Yes'. A simple flag for whether the passenger had family aboard.
Age Group	Splits Age into child (under 18), adult (18-59), and senior (60+), to see how survival differed across life stages.

Part 3: Exploratory Data Analysis

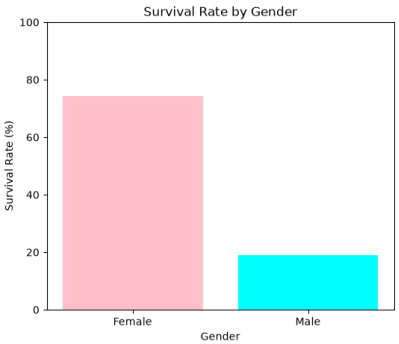
Q1: What percentage of passengers survived?

I used a pie chart because we are comparing two values as percentages, and a pie chart is best suited for that. 38.4% of passengers survived and 61.6% did not, shown in different colors.



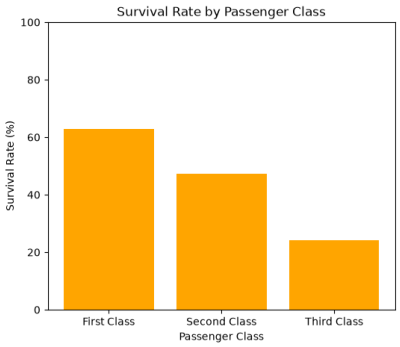
Q2: Which gender had the highest survival rate?

Females survived at more than 75%, while males only survived around 20% — meaning women and children were evacuated first.



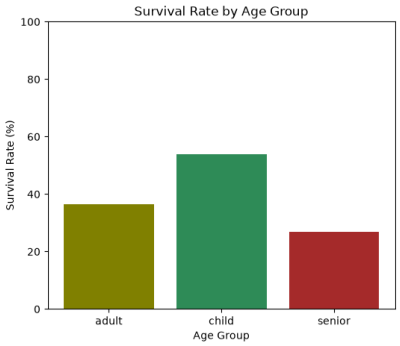
Q3: Which passenger class had the highest survival rate?

First Class had a higher survival rate than Second Class, which was higher than Third Class. Upper class passengers were likely evacuated first, possibly due to the placement of their rooms.



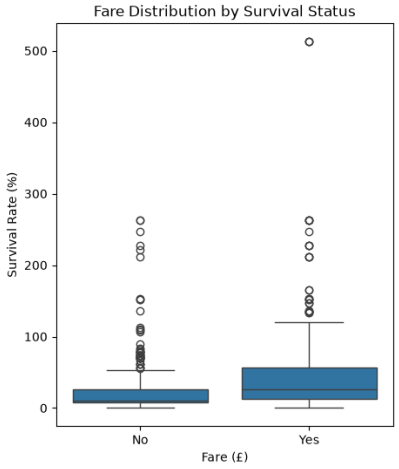
Q4: Does age affect survival?

Children survived the most, followed by adults, then seniors. Children were evacuated first, adults survived thanks to their strength, and seniors had the lowest survival rate as they were physically weaker.



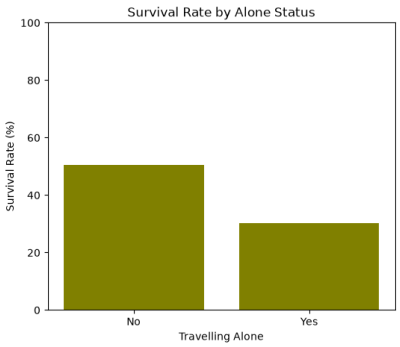
Q5: Does fare influence survival?

Survivors paid more on average — a median fare of £26 versus £10.50 for those who didn't make it. This matches the class result above: fare is essentially a proxy for passenger class, and class was clearly related to who got off the ship.



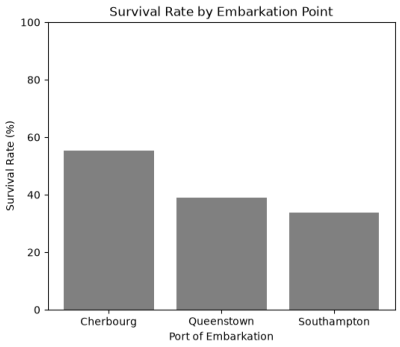
Q6: Did passengers travelling alone survive more often?

No — passengers travelling alone did not have a higher survival rate. Family members likely helped each other reach safety, while solo travellers had no one helping them.



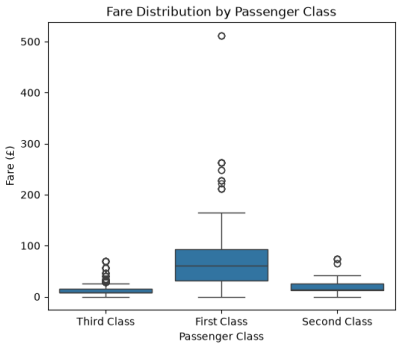
Q7: Which embarkation port had the highest survival rate?

Cherbourg passengers survived the most, followed by Queenstown, then Southampton. This is mainly due to the class mix from Question 3 — the more First Class passengers a port had, the higher its survival rate.



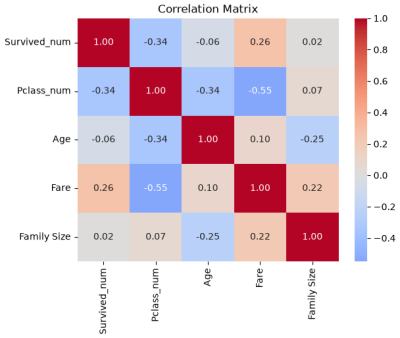
Q8: What is the relationship between passenger class and fare?

The higher the fare, the better the class, and the better the survival rate.



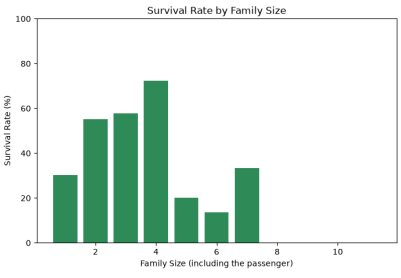
Q9: Correlation Heatmap

Fare has the strongest positive link with survival (0.26) — passengers who paid more survived more often. Passenger Class and Fare have the strongest relationship on the grid at -0.55, confirming a better class meant a pricier ticket. Age barely correlates with anything.



Q10: Extra Analysis — Family Size vs Survival

Like Question 6, families of 2-4 had a higher survival rate than solo travellers, likely because they helped each other. But families of 5-6 had a lower survival rate, likely due to the difficulty of coordinating a larger group.



Part 4: Statistical Analysis

Column	Mean	Median	Mode	Std Dev
Age	29.36	28.00	28.0	13.02
Fare	£32.20	£14.45	£8.05	£49.69
Family Size	1.90	1.00	1	1.61

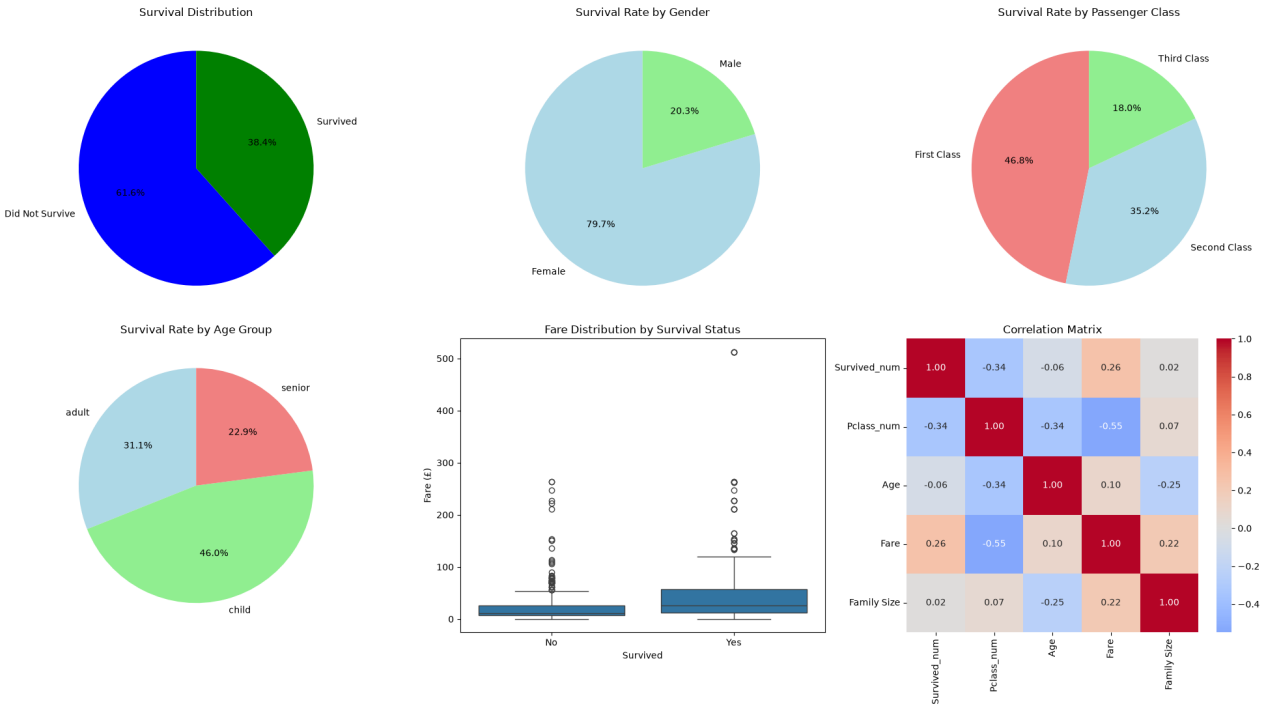
Age: Mean and median are close together, meaning the distribution is fairly symmetric with no extreme skew. The standard deviation is moderate, so most passengers fall within about 13 years of the average.

Fare: The mean is more than double the median, and the standard deviation is even bigger than the mean — a clear sign of heavy right-skew, caused by a small number of expensive First Class tickets pulling the average up.

Family Size: The mode and median are both low, since most passengers travelled alone. The mean sits a bit higher, pulled up by the smaller number of larger families aboard.

Dashboard

The six charts below bring together the survival, gender, class, age, and fare patterns, plus the correlation heatmap, in one view.



Business Insights

1. Only 38.5% of passengers survived while the other 61.6% did not.

(Additional insights to be added.)